

QUESTIONS

Part 1: Intermediate (Impact & Scale)

Focused on understanding the overall impact and distribution of the pandemic.

1. The Survival Metric

What is the Case Fatality Rate (CFR) for each country, expressed as the percentage of deaths relative to confirmed cases? Rank countries from highest to lowest.

2. Population Penetration

Which 10 countries experienced the highest infection rates relative to their population size?

3. Regional Contribution

For countries with multiple administrative regions (such as provinces or states), which region contributed the largest share of the country's total deaths?

4. The "First 100" Benchmark

How long did it take each country to progress from its first reported case to reaching 100 total deaths?

5. Global Hotspot Center

Where is the geographic center of the most severe outbreak zones, considering the top 5% of global hotspots weighted by case counts?

6. Recovery Lag

At a global level, what is the average time delay between surges in confirmed cases and subsequent increases in recoveries?

7. Data Integrity Audit

Which countries exhibit inconsistencies in reported data, where the sum of recoveries and deaths exceeds total confirmed cases? What is the total magnitude of these discrepancies?

Part 2: Advanced (Momentum & Trends)

Focused on understanding trends, timing, and rate of change.

8. Trend Smoothing

What is the 7-day moving average of new cases for key countries such as the United States and India, to account for reporting fluctuations?

9. Peak Momentum Identification

On which date did each country experience its highest number of new daily cases? Was there a period when multiple countries peaked simultaneously, indicating a global surge?

10. Growth Acceleration (Doubling Rate)

During the early phase of the pandemic, how quickly did total cases double in size across different continents?

11. Month-over-Month Growth

How did infection levels change month over month during 2020? Which month experienced the highest rate of growth?

12. Sustained Decline Analysis

Which countries managed to achieve a continuous period of at least 14 days of declining new cases, indicating effective control over the spread?